

Bachelor thesis CSAI

Please note:

This document is retrieved from the Canvas Bachelor Thesis Course Page

General thesis information

What is this course about?

In this course, you will conduct a supervised empirical quantitative research study in cognitive science or artificial intelligence, where you will tackle problems with a blend of curiosity, creativity, and analytical thinking. This experience will allow you to gain comprehensive insight into all phases of research, effectively applying the knowledge and skills you've acquired throughout your studies. Additionally, you will develop an in-depth understanding of a specific application area, which will serve as the theoretical foundation of your thesis.

Requirements

To be eligible for admission to the bachelor's thesis, all courses of the propaedeutic phase must have been successfully completed, and at least 60 ECTS from the post-propaedeutic phase of the bachelor's program must have been obtained.

Course Learning Goals

After completing the bachelor thesis project, the students will have obtained experience with all the phases of empirical quantitative research in cognitive science and artificial intelligence. They will have applied the knowledge and skills acquired in different courses.

On successful completion of this course, students will be able to:

1. Carry out scientific research in the combined domain of cognitive science and artificial intelligence under supervision and evaluate its merits.
2. Create programming code for the purpose of conducting experiments and/or data analysis, in line with best practices and coding standards.
3. Report orally on the scientific and social-cultural merit of their project.
4. Write an academic report on the research in line with academic standards.
5. Critically reflect on the conducted scientific research and the process.

Study Load

The Bachelorthesis CSAI consists of 12 ECTS; this corresponds to $12 * 28 = 336$ study hours. Apart from the weekly supervision sessions, the majority of these study hours will be

dedicated to self-study: working on your research project, preparing the deliverables and writing your thesis.

Deliverables

1. Project Proposal

- **Description:** This is the first opportunity students have to present their project.
- **Format:** A well-thought-out individual **written** proposal demonstrating the feasibility of the project. The proposal must include a literature review of scientific articles relevant to the content of the project.
- **Assessment:** Pass/Fail (**mandatory**).

For more details see [Project Proposal Guidelines](#) in Modules.

2. Poster Presentation

- **Description:** This is the second opportunity students have to present their project to scientific staff members and fellow students in an exhibition-like setting.
- **Format:** A short oral presentation (max 5 minutes) of your poster (to be submitted on Canvas) during the poster presentation session.
 - Upon reasonable request, alternative presentation settings can be arranged. Please contact your Thesis Coordinator.
- **Assessment:** Pass/Fail (**mandatory**).

For more details see [Poster Presentation Guidelines](#) in Modules.

3. Final Deliverables

3.1 Thesis Report (written): more detailed information can be found in the [Thesis Report Guidelines](#).

- **Assessment:** numerical grade attached

3.2 Programming Code: created for the project (e.g., R code for statistical analyses and visualizations, Python code for machine learning, Unity code for VR applications).

- **Assessment:** Pass/Fail (**mandatory**).

3.3 Self-Reflection Essay regarding a reflection on the student's development.

- **Assessment:** Pass/Fail (**mandatory**).
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Assessment and Grading

All deliverables for the thesis course are **mandatory**. The thesis report is graded **numerically**, while the project proposal, poster presentation, programming code, and self-reflection essay are assessed on a **Pass/Fail** basis: to pass each criterion in the rubric must be met with a passing grade.

Note on fraud/plagiarism - Students are fully responsible for the scientific integrity of their submitted work and must be fully transparent in disclosing any sources and technology used in their work. For more information, students are referred to the [Plagiarism FAQ](#).

Detailed Grading Process:

1. Project Proposal and Poster Presentation:

- - **Project Proposal:** The individual proposal may be discussed during supervision before the grade is released, indicating the likelihood of passing or failing.
 - **Poster Presentation:** The poster presentation will be evaluated by the supervisor, or their substitute. In the case of a potential fail, a second assessor will be involved.

Feedback: If applicable, students should integrate the feedback given by their thesis supervisor, other lecturers, and peers in the final deliverables to be submitted at the end of the project.

The proposal and presentation are assessed independently in terms of timeline; the proposal grade might not be determined before the presentation event.

Go / No Go: Students must pass both the proposal and poster presentation to proceed to the final deliverables submission. Failing either will halt the thesis process.

2. Final Deliverables

- **Assessment Criteria:**
 - **Pass/Fail Deliverables:** To pass each criterion in the rubric must be met with a passing grade.
 - **Thesis Report:** Evaluated by both the first reader (thesis supervisor) and the second reader (independent examiner). Each component must receive a passing grade (6 or more) from both examiners.
- **Final Grade Calculation:**
 - The final grade for the thesis report is determined by the thesis report grade.
 - The final grade is calculated by averaging the evaluations from both readers, provided all components receive a passing grade from both readers (6 or more).
 - The first reader provides the joint grade for the thesis, which will appear on Canvas.

Resubmission Process:

- If any deliverable fails, only the failed components must be resubmitted.
- The higher grade among two submissions counts.

- If substantial changes are made to the code for the second submission, resubmission of the Programming Code is recommended. The passing grade will be retained.
- Grades and feedback for both submission opportunities are released simultaneously on the predetermined date.

Communication and Recording of Grades:

- The final deliverables are assessed in line with the Exam Rules and Regulations.
- The first examiner is responsible for the judgment and quality of the assessment, while the second examiner ensures fairness.
- The thesis coordinator delivers the results to the Educational Office, where they are recorded in the TiU education information system (OSIRIS).
- The final verdict is communicated to the student within 15 working days after submission.

Display of Grades:

- The thesis grade will be in Osiris on a 10-point scale. Canvas displays the component sums, and a sum over 60 is displayed. It does not automatically round the grade to a failing grade represented by 30 if a criterion fails. If any criterion or deliverable receives a failing grade, the thesis fails (represented by 5).

A passing grade in all deliverables is required to complete the thesis course.

Timeline

- **Regular Supervision Meetings:** End with the first submission deadline.
 - Students are given **grade justifications** for their **first attempt** and can use this feedback to **improve their work for the second submission**.
- **Strict Deadlines:** To ensure fairness, no deviations are granted, unless valid reasons are provided.
 - Contact your study advisor ([Suzanne Warmerdam](#) or [Simone Willekens](#)) as soon as possible for personal circumstances causing delays.

For an overview of the schedule, see the calendar below:

Calendar week	Date	Details
35 / 2025	August 25	Start of supervision
38 / 2025	September 15	Assignment: Thesis proposal DRAFT
40 / 2025	September 29	Assignment: Thesis proposal

43 / 2025	October 20	Assignment: Thesis proposal RESIT
43 / 2025	October 23	Assignment: Poster presentation sessions In-person event in Dante Foyer from 13:00-15:00. Two 1-hour sessions — schedule TBD.
44 / 2025	October 27	Final go/no go decision
49 / 2025	December 1	Assignment: First submission deadline
51 / 2025	December 19	Grades first submission available
3 / 2026	January 12	Assignment: Second submission deadline
5 / 2025	January 30	Grades second submission available

Proposal guidelines

Last edit: August 21, 2025

General Information

The project proposal is your first opportunity to present your project and its deliverables, including a literature review of scientific articles relevant to your project. This proposal is evaluated on a pass/fail basis and serves as a summative assessment.

Your proposal will be reviewed by your intended supervisor and, if necessary, a second reader. Evaluation criteria include clarity (is it clear what will be done in this project?) and completeness (is the proposal well-thought-out and does it address all important aspects of the proposed project?). Proposals with spelling, grammar, and style mistakes will not be evaluated; you will be asked to resubmit a corrected version (this is referred to as a "technical re-submission" and does not count as a separate submission opportunity).

In the course of the semester, your project plan may need to be adapted as you learn more about the data. This is acceptable as long as the overall goal (the task you are addressing, the dataset you are using) remains generally the same. If your project undergoes significant changes from what was proposed, you need approval from your supervisor.

Format and Requirements

- **Length:** The project proposal should be a maximum of 4 well-written pages (1.5 line spacing, font size 11, 1 inch margins, DIN A4), excluding the title page, references, tables, and figures. No appendices are allowed.
- **Language and references:** The proposal must be written in correct Academic English and adhere to APA Style.
- **Title page:** On the title page, include your full name, email address, student number, the name of your internal supervisor, and the contact information of your external supervisor, if applicable.

Outline and Contents

Please study the rubric of the [assignment](#).

Project Definition

Provide a clear description of the problem you plan to address.

Motivation

Briefly explain why this problem is worth addressing from both societal and scientific perspectives.

Background

Summarize what is known in the scientific literature about this problem. Base this on at least 5 relevant sources that meet the following requirements:

1. **Recency:** Published in the last 5 years (exceptions may apply if recent publications are scarce; consult with your supervisor).
2. **Quality:** Published in scientific peer-reviewed journals or conference proceedings.
3. **Usefulness:** Should help frame the theoretical background of your project.

A full literature review is not required at this stage but will be expected in the final report.

Design Experiment or Dataset Description

Describe the design of your experiment or the dataset(s) you will use in as much detail as possible (size, format, accessibility). Provide a rationale for choosing these data.

Instrumentation and Material

For experimental studies, describe the instruments and materials you plan to use.

Statistics, Algorithms, and Software

Describe the statistical analyses or data mining and machine learning algorithms and software you plan to use. For data mining/machine learning algorithms, explain how you will evaluate your results. For prediction problems (classification or regression), outline the baseline methods you will compare your algorithms against and how you plan to obtain ground-truth labeled data to measure metrics like accuracy, precision, and recall. If using unsupervised techniques, explain how the clustering algorithm will be tested.

Milestones and Plan

Outline the major intermediate milestones and provide a general idea of your planning (this can be in the form of a table).

Avoiding Plagiarism

Ensure you do not commit plagiarism, which is considered a serious fraud. Suspected cases will be reported to the Examination Board, and established fraud will result in the assignment being declared invalid. In the case of a thesis, a new thesis will have to be written. Refer to Article 16 of the Rules and Guidelines for TSHD for the procedure and consequences of fraud. Tilburg University defines plagiarism as using parts of a text written by someone else, or the reasoning or ideas of others, without due acknowledgment. For more details, visit [Tilburg University's plagiarism guidelines](#)

[Links to an external site.](#) and the [Plagiarism FAQ](#).

Useful Resources

- [TSHD Education and Examination Regulations \(EER\), including the Rules and Regulations](#)

Thesis Report Guidelines

Last updated on August 13 2025

The guidelines below outline the content requirements for your thesis report. Please consult with you supervisor if you have any questions about these guidelines.

Disclaimer: Please be aware that these guidelines have an advisory function; see the [Bachelor thesis written report rubric](#) for the formal criteria on which the grade for your thesis is based.

Objectives

The Cognitive Science and Artificial Intelligence bachelor projects aim for students to conduct supervised quantitative research in cognitive science or artificial intelligence. Students should approach their projects with curiosity, creativity, and analytical thinking. They will develop in-depth knowledge about a specific application area discussed in the theoretical background of their thesis.

Copyright and collaboration

Make sure your thesis does not contain unacknowledged quotes or paraphrases as these could be detected as cases of plagiarism. For the same reason, do not reuse any figures or other images without a proper license or a permission of the author; if you have the permission, the author still needs to be credited. If you collaborated with someone else on some part of the project, indicate clearly which part of the work you are building upon was done by someone else.

See also the [Plagiarism FAQ](#) for more information.

Use and declaration of AI and AI-assisted technologies

Students must disclose the use of AI and AI-assisted technologies in the writing process in the Data Source, Ethics, Code, and Technology (DSECT) Statement (see instructions below).

Written Report Content Requirements

Length:

- The manuscript should have a maximum of 8,000 words, excluding references and appendices.

Elements of Your Report:

1. **Title Page:**

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- Contains the title, author, and other standard information (also see the [template](#))
- The title is:
 - clear (avoid ambiguity), concise, informative, and precise (avoid being overly general or vague).
 - less than 15 words long.

2. Abstract:

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- A summary of your thesis (150-250 words) addressing the following points:
 - What problem is being investigated?
 - What is your research goal? It should follow from how other researchers in the past addressed the problem (i.e., in terms of approach, focus, etc.).
 - What distinguishes your approach from theirs?
 - What are the essential features of your method?
 - What dataset are you using?
 - What are the main findings

3. Data Source, Ethics, Code, and Technology (DSECT) Statement

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- Provide answers to all below questions for transparency purposes when writing your ethical statement. Keep in mind, however that there is no one-fits-all solution to ethical statements, and they heavily depend on the nature of the study.
- Data Source
- - - Who is the owner of the data?
 - Does the thesis project involve collecting data from human participants or animals?
 - Does the owner of the data give consent?
 - How and through what channel is the data acquired?
- Figures
- - - Did you create all the images and figures?
 - Do you have consent for the images that you did not generate?
- Code
- - - Did you use parts of the code from another study/ someone else?
 - Did you list, including version number, all libraries and frameworks used?
- Technology

The following statement should be included: During the preparation of this work the author(s) used [NAME TOOL / SERVICE] in order to [REASON]. After using this

tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the thesis.

- - - Did you use any tools or services to paraphrase the given text (for example a thesaurus or the Academic Phrasebank)? Please name
 - Did you use any tools or services to check spelling or grammar? Please name them.
- - - Did you use any tools or service to typeset the given text? Please name.
 - Did you use any reference management software other than the latex template? If so, please name it in your statement clearly
 - Did you use any generative language models to help with your writing? Please name them and explain how you used these models.

4. Section 1: Introduction

- - This section mainly relates to the "Formulation goals, framework project" criterion of the rubric.
 - Explain briefly your research goals, why they are important, how you approached them, and what your findings were.
 - By reading the introduction, the goals of the project or research questions and the framework of the project (as in the research strategy / what the proposed approach is) must be clear to the reader. This approach should be motivated such that the reader can see how it fits the objective of the research.
 - Context: Start with the goal of your research.
 - Describe the context of your thesis in a very concise manner. Explain briefly the research domain, what the state-of-the-art is, and why the subject matter is interesting. Your readers should be left feeling that your thesis deals with an issue that is both important and interesting.
 - Explain the scientific and social relevance of your work.
 - You do not need to go to great lengths to describe every relevant prior study yet; that is reserved for the Related Work section. For now, it is fine to state something along the lines of: "This issue has been addressed extensively (see the Discussion section)".
 - Devote one paragraph of the introduction explicitly to the scientific relevance of your project. Note that scientific relevance could be derived from the domain-specific research goal addressed in your research, or in the proposal of a new algorithm or approach.
 - Explain societal relevance by linking you research goals and potential outcomes to applications and/or insights that benefit society (i.e., beyond the scientific community).
 - Research questions/goals: Once the context is established, specify your research questions/goals.
 - Findings: Give a one-paragraph overview of your main findings.

5. Section 2: Related Work

- - This section mainly relates to the "Theoretical underpinnings, use of literature" criterion of the rubric.
 - This section includes references to relevant theories and models, which should be related to the description of the project at hand.
 - Avoid redundant references here.
 - The discussion of the literature should allow the reader to put the work into perspective. There should be a comparison in terms of this work versus the papers cited, for similarities and differences.
 - Include existing recent literature to provide a good state-of-the-art overview.
 - Explain the larger scientific context of the problem: what is the theory behind it, if any, what previous research has been done related to it, and how your work builds on this related research.

Below are some step-by-step guidelines for writing this section:

- - Specify the area of research in which your work belongs and provide a context for the research focus. What research issue is your work focused on? Why is it an issue of importance?
 - Aim to cite five sources from the last five years. If this domain is not active in recent years, and there are no recent papers, note that in the thesis.
 - Describe relevant work conducted in the same research area (with proper references). Has this issue been addressed in the literature? By whom? What have they done and found? What are the relevant theories? Are there any contradicting findings or competing models/theories? What is the state of the art?
 - Identify research gaps and/or shortcomings of existing method; define research problems:
- - - What is missing from prior research? What are the limitations of existing models? Could there be alternative approaches to solving the same problems?
 - Specifically, what research problems are left unanswered? What insights or implications will you tackling these research problems bring about?
 - Specify the research questions and goals of the current work; announce the methodology you are going to adopt. What are the research goals? What are you trying to achieve with the current work? How are you going to fill the research gaps? What sets your work apart from prior research? What dataset are you going to use?

Common problems and remedies:

X Failure to maintain focus on the research goal, by including references to studies that are only remotely related to yours.

✓ Make sure you are not trying to impress the readers by the broad scope of your knowledge, thereby forgetting that they are interested in your current research only.

X Failure to support statements with adequate references.

✓ Always give credit where credit is due. If you are making a statement along the lines of: It has been established in prior research that..., make sure you follow the statement with references.

✗ Failure to express arguments or ideas in your own words.

✗ It is not acceptable to simply paraphrase the work of someone else by changing a few words here and there, without acknowledging the source. Failure to do so is a case of plagiarism and can lead to severe consequences!

✓ If you must include a direct quote, enclose it with quotation marks and specify the page number in your reference.

✗ Failure to include references to recent work.

✓ Whilst certain dated works remain important and are still widely cited (e.g., Gold, 1967, if the research concerns grammatical inference), try to stay on top of developments in the field and refer to the more recent literature, for example include references from the last five years in the field.

✗ Discussion of related work is not relevant or appropriate for the research project at hand.

✓ Make sure that the discussion of related work is explicitly linked to the research question and methods of your research project. Do not include unnecessary detail.

✗ Failure to critically reflect on the literature.

✓ Demonstrate awareness of relations among existing models or studies by specifying any relevant commonality, contradiction, or inconsistency among them.

✗ Failure to give a convincing rationale for conducting the current study.

✓ Explain how the current work continues and improves upon previous lines of enquiry. Be explicit about the contribution of the current work

6. Section 3: Methods

- - This section relates to the "Use of methods and processing data" criterion of the rubric.
 - In this section, you describe your general approach, for example, which experimental methods, mathematical models, or computational algorithms you used.
 - Description of methods and analysis of information/data should be complete (for reproducibility).
 - For experimental projects, describe experimental design, including dependent and independent variables and how they are measured.
 - For machine learning projects, explain which models and data are used, how data is preprocessed, and how models are evaluated.

- Used methods and analysis of data/information should be appropriate for the goals of the project.
- The chosen methods should be compatible with the research goal and this should be explained clearly.
- Notation should be clear and consistent. It's not necessary to elaborate in detail on standard or common methods/techniques.
- This section often contains a combination of mathematical formulas, diagrams, and verbal descriptions.

The following information should be covered:

- - Description of the experimental procedure (if relevant):
 - Was it research done on humans or simulations?
 - How was the population recruited and the test administered?
 - What was the task that the participants completed?
 - In the case of simulations, which algorithm was used, which parameters were chosen and how?
 - Description of your dataset (if relevant): the organization offering the dataset, sample size, how and when the data was collected, which features could be found in the data, and any other relevant information
 - Where appropriate, report descriptive statistics to offer a better impression of the dataset.
 - Cleaning / pre-processing of data: was there any oddity (e.g., error) in the dataset and what was done about it, which parts of the data were discarded and why, whether or not certain features were transformed and why, what was done about the missing values and why, and any other preprocessing done.
 - Always justify your decisions with theoretical and/or statistical arguments.
 - Description of statistical tests / machine learning models that are applied in the research.
 - Description of the actual implementation, i.e., programming languages and versions, packages used, proprietary applications supporting the coding, cross-validation methods, etc.
 - Description of evaluation criteria (if appropriate).

Common problems and remedies:

X Symbols in formulas are not defined or explained.

✓ Make sure that it's clear what the notation stands for.

X Failure to list all important details.

✓ Always write with other researchers in mind and include all relevant details. When in doubt, ask yourself: If I were to leave this piece of information out, would other researchers be able to reproduce my work?

X Failure to justify choices made.

✓ Always be explicit about the rationale for making certain choices; they should be made on theoretical (e.g., prior research), methodological (e.g., algorithmic bias) or empirical grounds (i.e. tuning on validation data). For example, it is better to use only one or two algorithms properly than to try out every algorithm under the sun without proper justification and in a superficial way.

7. Section 4: Results

- - This section is related to the "results and reflection on results" criterion of the rubric.
 - Present your results in a structured manner, often with the help of tables or figures.
- - - In your text, do not simply restate the information listed in the tables or figures. Try to make sense of the results, highlighting important or interesting findings that you might revisit in the discussion section. The figures are not the presentation of your results but their illustration.
 - Do not leave information presented in tables or figures unexplained. You have included information there for a reason, so take the time to go through it (e.g., explain what each column is about).
 - Provide high-quality clear figures with well-sized legends and informative captions. All axes in a graph must be labeled.
 - Do not cluster tables and figures – there should always be some text in between.
 - Larger tables (with more than ca. 10-15 rows) should be placed in the appendix.
 - Where appropriate, explore the results further using statistical analysis, confusion matrices, or visualizations
 - The goal is to obtain a more fine-grained understanding of your results, uncovering patterns that might not be obvious from the overall results (for example, does the overall pattern of results hold across ages and genders? Or, in the case of an AI model, does the predictive performance of the model differ greatly between classes).
 - Make use of subsections and describe in one or two sentences what results will be reported in each subsection.
 - For example: In this section, classification performance for the feature types described in section 3 on X dataset will be presented.
 - Describe the model performance or statistical results, and make use of tables and/or figures.

Common problems and remedies:

✗ Failure to report the baseline performance for a machine learning problem.

✓ Always report the baseline performance, as it is difficult to interpret the results without knowing the basis for comparison (e.g., previous research, chance-level performance, etc.)

X Failure to interpret information listed in tables and figures.

✓ Elaborate on the takeaway of (your findings in) the tables such that your readers know what they should focus on.

X Failure to use the correct type of figure.

✓ Consult scholarly articles and books to see which type of figure is appropriate for which visualisation purpose.

X Failure to format numbers according to English-language conventions.

✓ Make sure you use decimal points, and commas as thousand separators (i.e. 1.2 and 10,000)

X Failure to properly reference tables and figures.

✓ Be clear about which tables or figures you are referring to. Make sure all tables and figures are referenced.

X Unreadable figures and graphs.

✓ Make sure all fonts are readable on a regular A4 sized print-out of your thesis and make sure title, captions and labels are present (in line with APA style).

8. Section 5: Discussion

- - In the Discussion section you evaluate your results with regard to the research goals presented in the introduction. You should indicate all weaknesses in the results and weigh them relative to each other. Better alternatives for the methods used should also be discussed here. The following are some recommended elements:
 - Remind you readers what the goal of your study was.
 - Discuss the findings, preferably in the same order in which they were presented in the results section.
 - If a finding was surprising, you should offer reasonable speculations as to why this particular result was observed.
 - It could be the case that your results only partially answered your research goals due to limitations of the model or the data. Acknowledge these limitations, offer possible solutions, and defend the validity of your results.
 - Put your results in perspective by making links to the literature.
 - Make very clear what the contribution of your study is within the existing framework.
 - Discuss future research directions that can be identified based on the results of your project.

Common problems and remedies:

X Failure to provide context for the results

✓ The discussion section should be understandable when standing alone. It is important that you spell out your research questions or goals again, so that researchers who only read this section can still have a good idea of what your findings are.

9. Section 6: Conclusion

The Conclusion is a short section where you restate the research questions/goals and provide the answers to them by combining the results you obtained with a very brief summary of how they can be placed in the context of existing research. Here you also explain the implications of your work for the field, and identify which directions future research could take, on the basis of the contribution of your study.

10. Acknowledgements (optional)

In this brief section, you can acknowledge sources of funding, data, or anyone who helped you with your research.

11. References

Use an author-year citation style such as APA.

12. Appendices and Supplementary Materials

Appendices are appropriate for extra, non-essential visualizations, examples and analyses. It is strongly encouraged, whenever possible, to store data and source code in an online repository and refer to it in the manuscript. Github.com is a common choice. It is not ideal to include code in the thesis report.

[Checklist for Fluency of Language and Writing](#)

The below list mostly concerns the "Fluency of language and writing skills" criterion of the thesis

- Is the length within the specified limits?
- Are there any redundancies (irrelevant content must be removed)?
- Is the content presented in a logically sound way?
- Did you run a spell checker?
- Did you proofread for grammar and clarity?
- Did you use English-language conventions for number formatting (decimal point, comma thousand separator)?
- Did you round excessively precise numbers?
- Are all quotes and paraphrases from other texts properly referenced?
- Are all figures either your own or used by permission?
- Are the symbols used in formulas defined or otherwise explained?
- Does the thesis conform to the necessary formatting of a Tilburg University TSHD bachelor thesis document, including approved citation styles and the title page?
- Is there a DSECT statement?

Templates

Thesis templates for LaTeX and Word can be found in the corresponding [Templates module](#). The provided templates are consistent with the Tilburg University Bachelor Thesis style requirements

If you opt to write your thesis without one of these templates, you must be sure your thesis is consistent with these style requirements.

The template uses APA6- or IEEE-style citations (using APA6 or later is versions are accepted), a bibliography and some formatting elements (e.g. 2-column for IEEE).